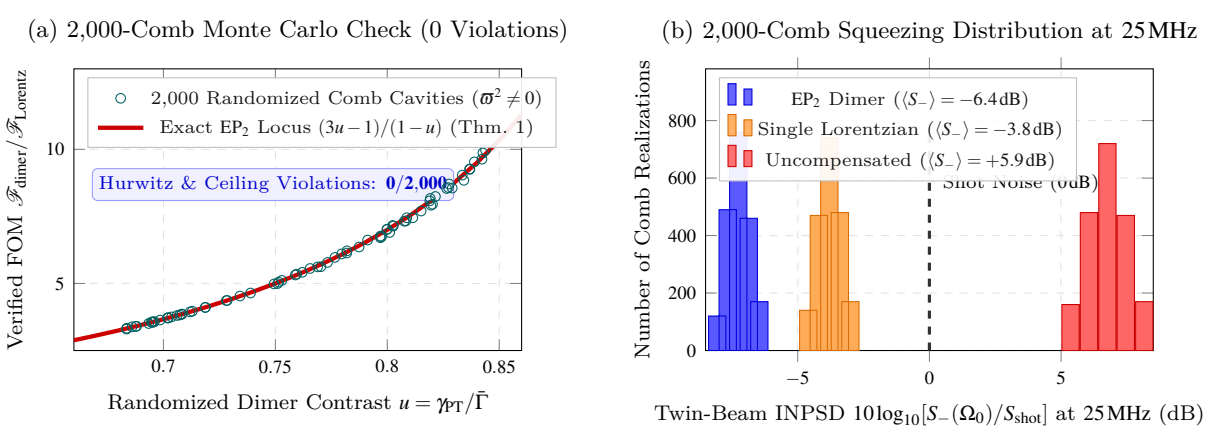
(a) 2,000-Comb Monte Carlo Check (0 Violations)
2,000 Randomized Comb Cavities ($\varpi^2 \neq 0$)
Exact EP$_2$ Locus $(3u-1)/(1-u)$ (Thm. 1)
Hurwitz & Ceiling Violations: 0/2,000
Verified FOM $\mathscr{F}_{\text{dimer}}/\mathscr{F}_{\text{Lorentz}}$
Randomized Dimer Contrast $u = \gamma_{\text{PT}}/\bar{\Gamma}$
(b) 2,000-Comb Squeezing Distribution at 25 MHz
EP$_2$ Dimer ($\langle S_- \rangle = -6.4$ dB)
Single Lorentzian ($\langle S_- \rangle = -3.8$ dB)
Uncompensated ($\langle S_- \rangle = +5.9$ dB)
Shot Noise (0 dB)
Number of Comb Realizations
Twin-Beam INPSD $10\log_{10}[S_-(\Omega_0)/S_{\text{shot}}]$ at 25 MHz (dB)